

MAT 322 HW07

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Problem 1. Let $U \subseteq \mathbb{R}^n$ be open and let $f: U \rightarrow \mathbb{R}$ be continuous. If f vanishes outside a compact subset $C \subseteq U$, show that $\int_U f$ and $\int_C f$ both exist and are equal.

Solution. Let $C_1 \subseteq C_2 \subseteq \dots \subseteq U$ be a sequence of compact subsets of U whose union is U . Let $C'_i = C_i \cup C$, so that now the sequence C'_i satisfies the same properties as C_i , but with each C'_i containing C . Now clearly

$$\int_{C'_i} f = \int_C f \leq \max_{x \in C} f \cdot \text{vol } C < \infty,$$

with existence of the integrals given by being nowhere discontinuous. Then

$$\int_U f = \lim_{i \rightarrow \infty} \int_{C'_i} f = \int_C f$$

as desired. $\square$

Problem 2. Let U be the region in $\mathbb{R}^2$ bounded by the curve $x^2 - xy + 2y^2 = 1$. Express the integral $\int_U xy$ as an integral over the unit ball in $\mathbb{R}^2$ centered at 0.

Solution. Let $a = \frac{\sqrt{2-\sqrt{2}}}{2}$ and $b = -\frac{\sqrt{2+\sqrt{2}}}{2}$, and let $s = \frac{3+\sqrt{2}}{2}$ and $t = \frac{3-\sqrt{2}}{2}$. One can then compute that the map

$$f(x, y) := \left(\frac{ax}{\sqrt{s}} - \frac{by}{\sqrt{t}}, \frac{bx}{\sqrt{s}} + \frac{ay}{\sqrt{t}} \right)$$

bijectionally sends the unit ball B centered at 0 to the region bounded by the ellipse $x^2 - xy + 2y^2 = 1$. Therefore, by the change of variables formula,

$$\begin{aligned} \int_{f(B)} xy &= \int_B g \circ f \cdot |\det(df)| \\ &= \int_B \left(\frac{abx^2}{s} + \frac{(a^2 - b^2)xy}{\sqrt{st}} - \frac{aby^2}{t} \right) \cdot \frac{a^2 + b^2}{\sqrt{st}} \\ &= \int_B \frac{4}{7} \cdot \left(\frac{\sqrt{2}}{2(3+\sqrt{2})} x^2 - \frac{2\sqrt{2}}{7} xy - \frac{\sqrt{2}}{2(3-\sqrt{2})} y^2 \right) \end{aligned}$$

where $g(x, y) = xy$. $\square$

Problem 3. (a) Express the volume of the solid in $\mathbb{R}^3$ bounded below by the surface $z = x^2 + 2y^2$ and above by the plane $z = 2x + 6y + 1$, as the integral of a suitable function over the unit ball in $\mathbb{R}^2$ centered at 0.

(b) Compute this volume.

Solution. The surface and plane intersect on the locus of points satisfying

$$x^2 - 2x + 2y^2 - 6y - 1 = 0 \iff (x-1)^2 + 2\left(y - \frac{3}{2}\right)^2 = \frac{13}{2}.$$

Let the region bounded by this ellipse be E ; we want to compute

$$\int_E 2x - x^2 + 6y - 2y^2 + 1.$$

Translate E to get ellipse E' centered at the origin first, so the integral becomes

$$\int_{E'} \frac{13}{2} - x^2 - 2y^2.$$

Then, take the map $(x, y) \mapsto (x\sqrt{2/13}, y\sqrt{4/13})$ to shrink E' to the unit disk, which gives

$$\int_{B^2(1)} \left(\frac{13}{2} - \frac{13x^2}{2} - \frac{13y^2}{2} \right) \cdot \frac{2\sqrt{2}}{13} = \int_{\theta=0}^{2\pi} \int_{r=0}^1 r^3 \sqrt{2} = \boxed{\frac{\pi\sqrt{2}}{2}}.$$

□

Problem 4. Let $B^n(R)$ denote the closed ball of radius R in $\mathbb{R}^n$.

- (a) Show that $\text{vol}(B^n(R)) = \lambda_n R^n$ for some constant λ_n .
- (b) Compute λ_1 and λ_2 .
- (c) Compute λ_n in terms of λ_{n-2} .
- (d) Obtain a formula for λ_n .

Solution. (a) Let λ_n be the volume of the unit ball $B^n(1)$. Then, we have

$$\int_{B^n(R)} 1 = \int_{B^n(1)} R^n = \lambda_n R^n$$

using the change of variables formula with the scaling-by-factor-of- R map.

(b) The volume of $B^1(1) = (-1, 1)$ is $\boxed{\lambda_1 = 2}$, and the volume of $B^2(1)$ is the area of the unit circle $\boxed{\lambda_2 = \pi}$.

(c) We can compute the volume by integrating the $n-2$ dimensional balls above each point in the x_1x_2 -plane: specifically, if one picks a point a distance of $r \leq 1$ from 0, then the cross-section of the sphere obtained by fixing x_1 and x_2 is $B^{n-2}(\sqrt{R^2 - r^2})$. Therefore, parametrizing the x_1x_2 -plane in polar coordinates, the volume of the n -dimensional unit ball is

$$\lambda_n = \int_{\theta=0}^{2\pi} \int_{r=0}^1 \lambda_{n-2} \sqrt{1 - r^2}^{n-2} \cdot r = 2\pi \lambda_{n-2} \cdot \left[-\frac{1}{n} (1 - r^2)^{n/2} \right]_{r=0}^1 = \frac{2\pi \lambda_{n-2}}{n}.$$

(d) If n is even, we get

$$\lambda_n = \frac{2\pi}{n} \cdot \lambda_{n-2} = \frac{2\pi}{n} \cdot \frac{2\pi}{n-2} = \dots = \frac{(2\pi)^{n/2}}{n!!}$$

and if n is odd,

$$\lambda_n = \frac{2\pi}{n} \cdot \lambda_{n-2} = \frac{2\pi}{n} \cdot \frac{2\pi}{n-2} = \dots = \frac{(2\pi)^{(n-1)/2}}{n!!} \cdot 2$$

where $n!!$ is the product of all positive integers less than or equal to n that have the same parity as n (so for example $5!! = 5 \cdot 3 \cdot 1$ and $4!! = 4 \cdot 2$). □